Labour Supply and Public Policy

ECON 300 - Labour Economics

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Winter 2026

Learning Objectives

- Discuss key features of demogrants, social assistance, unemployment insurance, wage subsidies, workers' compensation, and childcare subsidies.
- Use the labour supply model to analyze work incentives under demogrants, social assistance (with clawbacks), and wage subsidies.
- Analyze unemployment insurance within the income–leisure model and note empirical challenges.
- Graphically illustrate disability impacts on the budget constraint; compare compensation programs.
- Show how childcare costs and subsidies alter constraints and participation decisions.

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Main Questions

- Can Canadian transfer programs be analyzed in a common framework to assess "best" designs for work incentives?
- Does unemployment insurance necessarily reduce incentives to work?
- Can welfare be designed to minimize adverse incentives while targeting need?
- How do we incorporate disabilities into the labour supply model and evaluate workers' compensation?
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Income Maintenance Programs

- Aim to supplement low incomes that arise for varied reasons; no single program fits all causes.
- Distinguish **permanent** (low wage) from **transitory** (hours shock) shortfalls—policy tools differ.
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Universal vs. Targeted Programs

Universal Programs

Administratively simple; everyone receives the same transfer regardless of income. Expensive and benefits also flow to non-poor.

Targeted Programs

Cheaper: transfers top up to a threshold (e.g., poverty line) but can create **implicit taxes** on earnings that reduce work effort.

TABLE 3.1: Government transfers received by Census Families in Canada, 2017

Benefits	**Percent of families**	**Average Amount**
All government transfers (federal and provincial)	83.7%	$12,750
Federal Child Benefits	20.2%	$6,314
Provincial Child Benefits	10.6%	$1,941
Employment Insurance	14.0%	$8,352
Social Assistance	11.3%	$8,367
Workers' Compensation	3.5%	$9,820
Canada/Quebec Pension Plan	32.4%	$10,092
Old Age Security	25.1%	$8,566
Guaranteed Income Supplement	9.6%	$6,634

Demogrants

- Lump-sum transfer specific to a demographic group (e.g., OAS).
- **Work incentives:** No substitution effect; a pure **income effect** that tends to increase leisure and reduce hours.

Note: The income gain is partly “spent” on leisure, so income rises by less than the grant amount.

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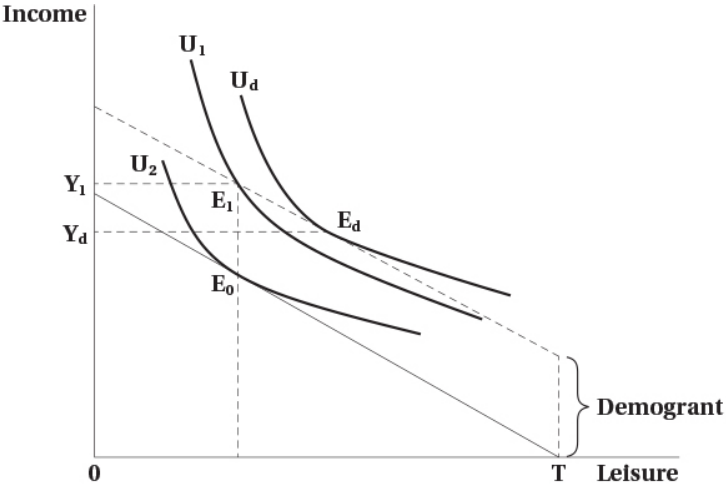
Figure 3.1: Work Incentive Effects of a Lump-Sum Demogrant



Welfare (Social Assistance)

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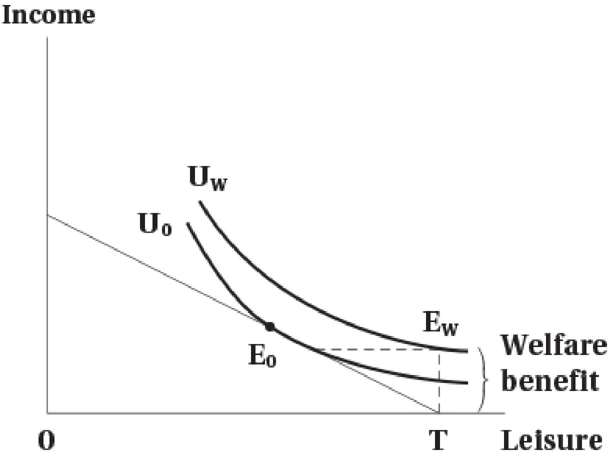
TABLE 3.2: Income of Welfare Recipients by Province, 2018 (lone parent, one child)

Province	Annual Income[a] ($)	Income as % of Poverty Line[b]
Newfoundland	23,436	85
Prince Edward Island	20,977	77
Nova Scotia	18,240	67
New Brunswick	19,978	78
Quebec	21,867	86
Ontario	21,463	72
Manitoba	21,764	82
Saskatchewan	21,087	77
Alberta	19,927	68
British Columbia	20,782	71
Average (unweighted)	**20,952**	**76**

a. Total annual income under program rules. *b.* Relative to poverty threshold used in the chapter.

Figure 3.2: Work Incentive Effects of a Welfare Benefit with 100% Clawback



Improving Work Incentives in Welfare

- Reduce the welfare benefit level (lower income effect).
- Raise the wage rate (tilt budget upward).
- Reduce the implicit tax rate (partial clawback).
- Shift preferences (e.g., work supports) toward market work.

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Figure 3.3: Other Work Incentive Effects of Welfare Programs

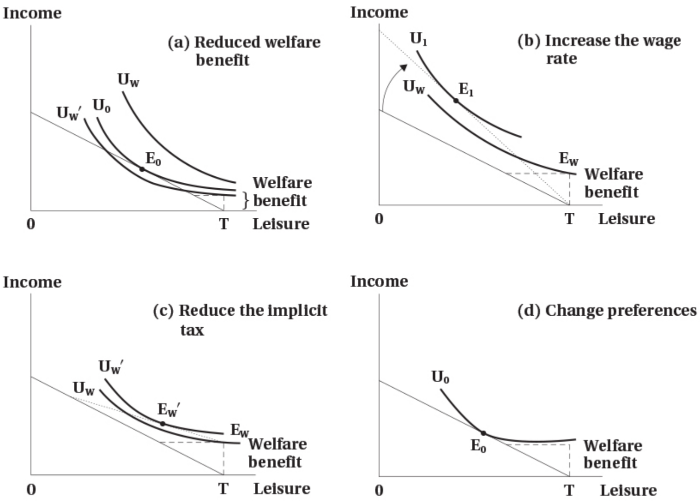
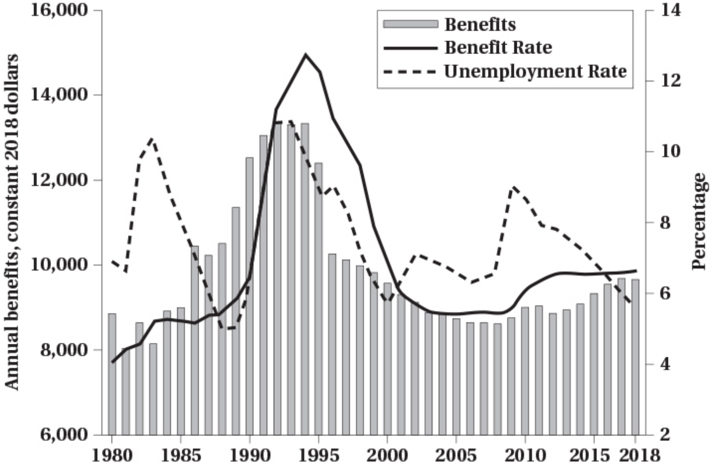


Figure 3.4: Welfare Benefits, Beneficiaries, and Labour Market Conditions (Ontario)



Negative Income Tax (NIT)

Income guarantee with partial clawback:

$$Y = G + (1 - t)E$$

- G: guaranteed income; t: implicit tax rate; E: earnings.
- Reduces hours via income effect and partial substitution effect (slope changes).
- Can replace welfare schemes with harsher disincentives.

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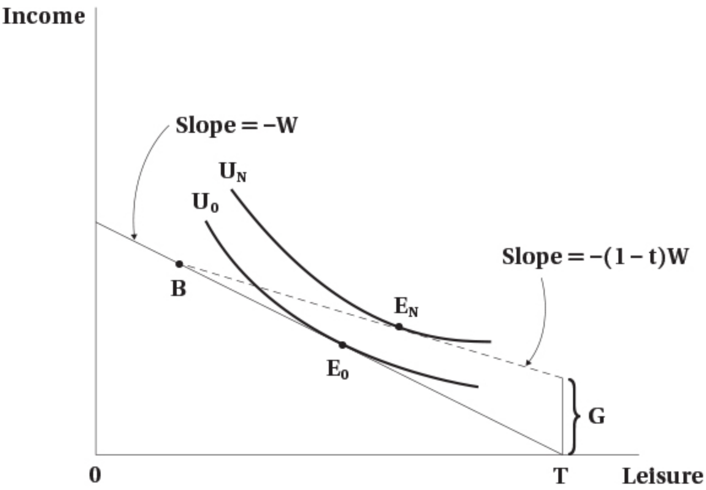

Wage Subsidy

- Subsidizes earnings; raises effective wage.
- **Theoretical net effect on hours is indeterminate** (income vs. substitution work in opposite directions).
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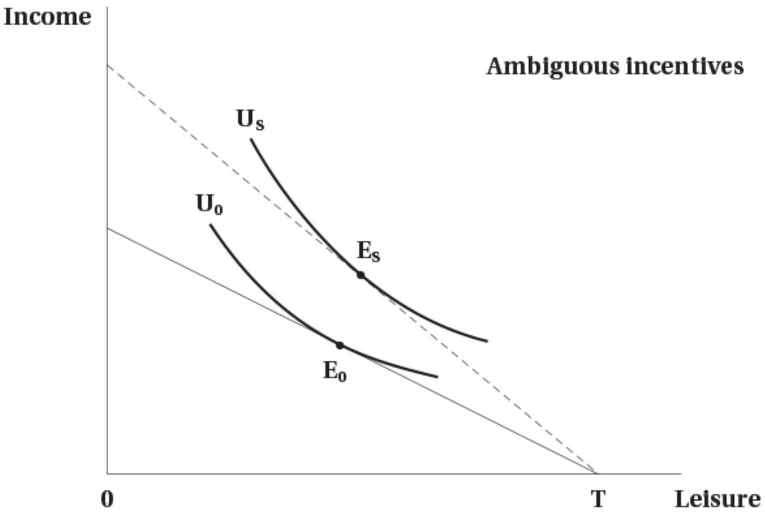

Refundable Tax Credit (Targeted Wage Subsidy)

- Income-tested wage subsidy: **phase-in**, **flat**, and **phase-out** regions.
- Phase-in: substitution $\uparrow$ (work more) vs. income $\uparrow$ (work less) $\Rightarrow$ ambiguous net.
- Flat: income effect only $\Rightarrow$ hours fall (if leisure is normal).
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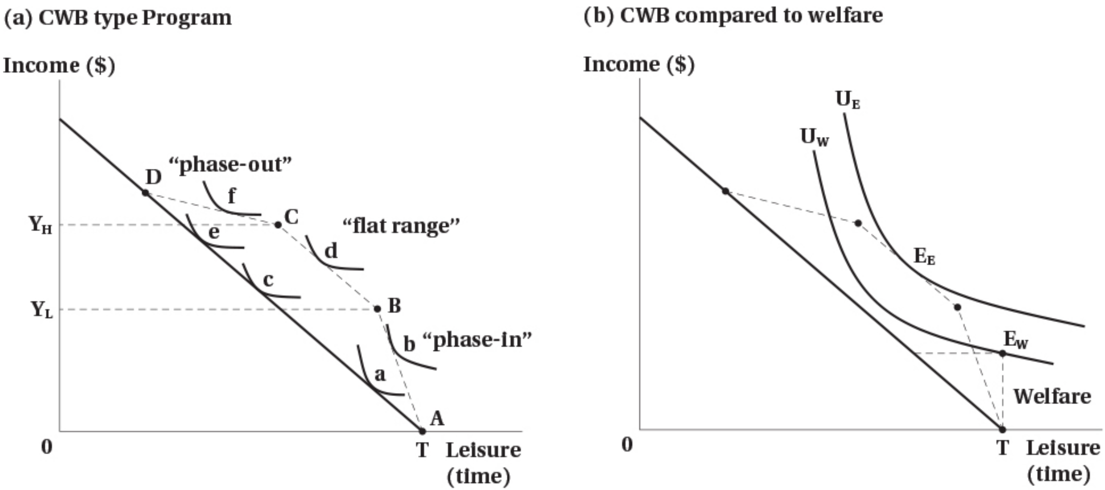

Unemployment Insurance (EI): Key Features

- Canada's unemployment insurance program, known as Employment Insurance EI since 1996, is the largest income security program for non-elderly individuals in the country
- The amount of income replacement rate is 55% of lost earning, subject to a maximum
- The duration of benefit ranges from 14 to 45 weeks, depending on the regional rate of unemployment
- In order to qualify for the benefits, individuals must have worked between 420 and 700 hours, the equivalent of 12-20 weeks at 35 hours per week, again depending on the unemployment rate of the region.

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Worked Example: Unemployment Insurance and Work Incentives

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Analysing the work incentive effects of unemployment insurance using the income-leisure model of the labour supply

- Recipients receive 60 percent of their weekly pay in the case of unemployment
- It requires a minimum of 14 weeks of employment in order to qualify for benefits.
- The maximum period of receiving benefits is 20 weeks
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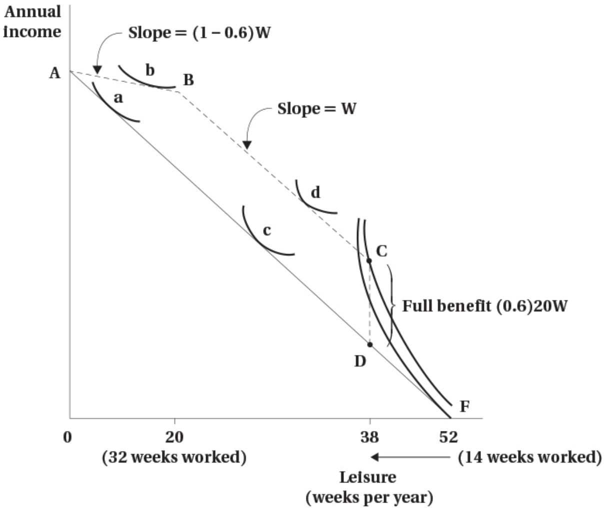

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- AF (solid line)- potential income constraint in the absence of unemployment insurance (UI) program
- ABCDF- constraint with UI
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- If the individual works 14 weeks, they are eligible for benefits for an additional 20 weeks
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- Their annual income would be $Y = 14W + (0.60)20W$

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The work incentive effects of UI depend on where the individual would be located in the absence of UI system - three different possibilities

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Income Constraint for a Simplified Unemployment Insurance Scheme

- Their annual income would be $Y = 14W + (0.60)20W$
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The work incentive effects of UI depend on where the individual would be located in the absence of UI system - three different possibilities

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Income Constraint for a Simplified Unemployment Insurance Scheme

3. Individuals who would be outside the labour force in the absence of UI program
   - originally locted at point F - reflecting the high value of their time devoted to non-market activities
   - many of these individuals neither work nor collect UI
   - others would enter the labour force and work a sufficient number of weeks to qualify for UI benefits (point C)
     - UI program has increased work incentives by making it attractive for these individuals to devote less time to non-market activities and to enter the labour force for relatively brief periods.

Effect of a Disability

- Alters budget (medical costs, fewer feasible hours) and/or preferences (higher disutility of work).
- Consider hours able to work, medical expenses, reduced wage-earning capacity, and relative disutility of market work.

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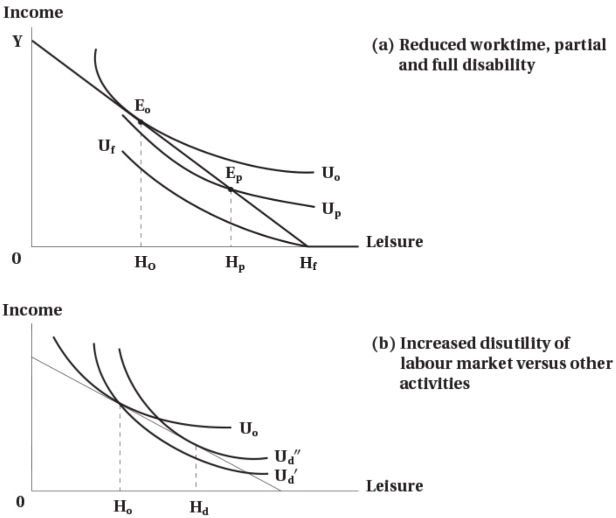
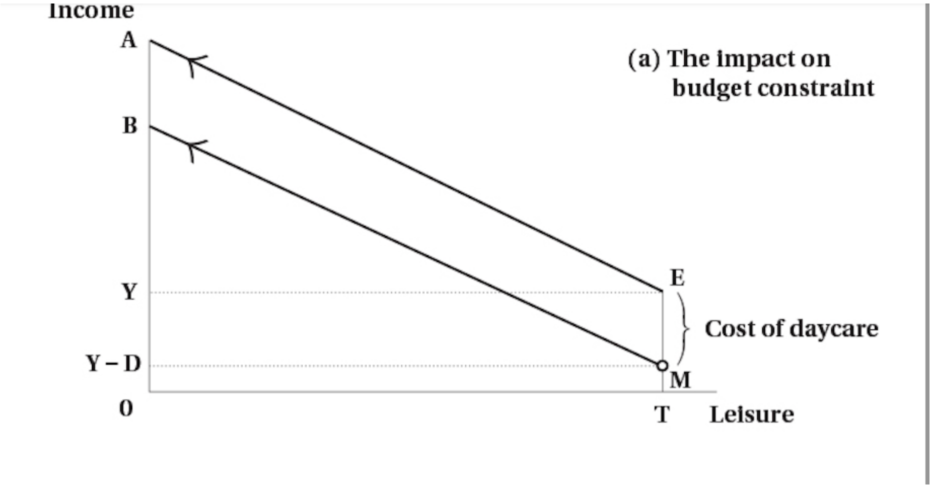
Figure 3.8: Effect of Disability



Child Care Benefit

- Analysis of the child care benefit first requires an investigation of how child care costs might affect a parent's labour supply decision
- Child care costs can be modelled as a special case of fixed costs associated with working
- Assumptions:
  - child care cost is incurred only if the person works in paid labour market
  - child care cost is independent of hours worked

The effect of Child Care Costs on Labour Supply



The effect of Child Care Costs on Labour Supply

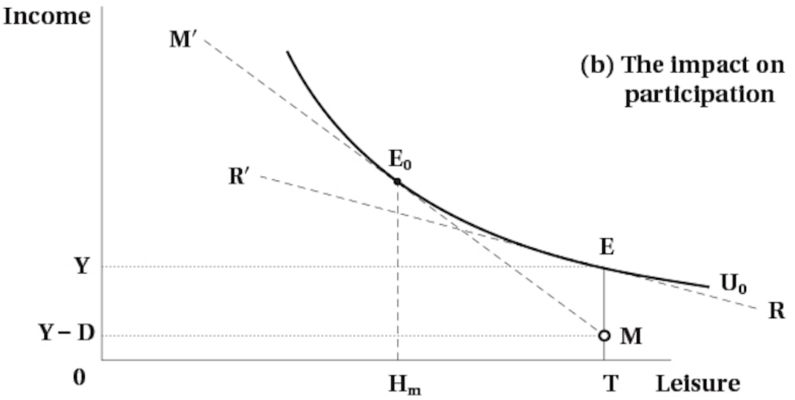
The impact on budget constraint:

- the fixed cost of child care is analogous to a vertical drop in the individual's budget constraint immediately upon entering the labour market
  - the individual incurs the fixed costs D=EM $\rightarrow$ analogous to a drop in income of EM
  - point M is slightly to the left of the vertical line ET - indicating that costs are incurred only if one engages in labour market activity
  - fixed costs are avoided if one does not work in the labour market but remains at E

Summary:

- Income constraint AE - in the absence of child care costs
- Income constraint BME - with fixed child care costs

The effect of Child Care Costs on Labour Supply



The effect of Child Care Costs on Labour Supply

The impact on participation:

- In the absence of child care fixed costs, the individual's reservation wage is depicted by the slope RR'

- In the presence of child care fixed costs, the individual's reservation wage is depicted by the slope MM'

  - the minimum wage above which one would enter the labour market

  - at this wage one would be indifferent between engaging in labour market activities at E_0 and remaining at home with children at E

- Market wage greater than the reservation wage would induce labour force participation and the amount of work would be determined by the tangency of new, higher indifference curve and a higher budget constraint

The effect of Child Care Costs on Labour Supply

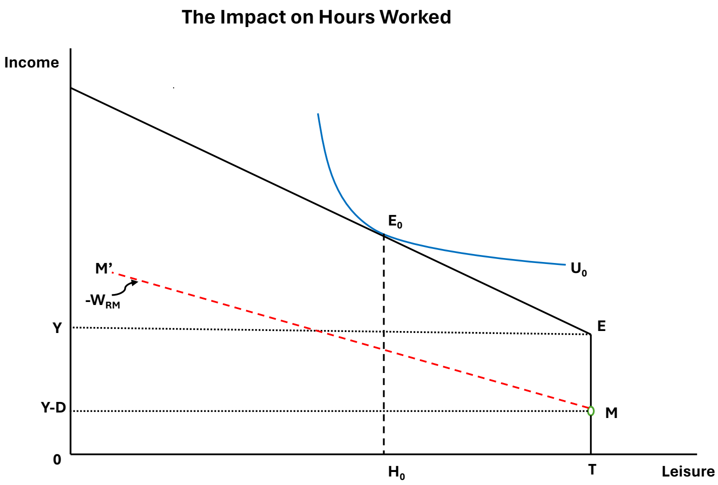
The impact on participation:

- the line MM' is tangent to U_0 at the point E_0 that is to the left of E of the same indifference curve

- the indifference curve become steeper as one moves to the left of E because of the hypothesis of diminishing marginal rate of substitution

- child care costs associated with entering the labour market increase the reservation wage by making labour market work less attractive than caring for the children at home

The effect of Child Care Costs on Labour Supply



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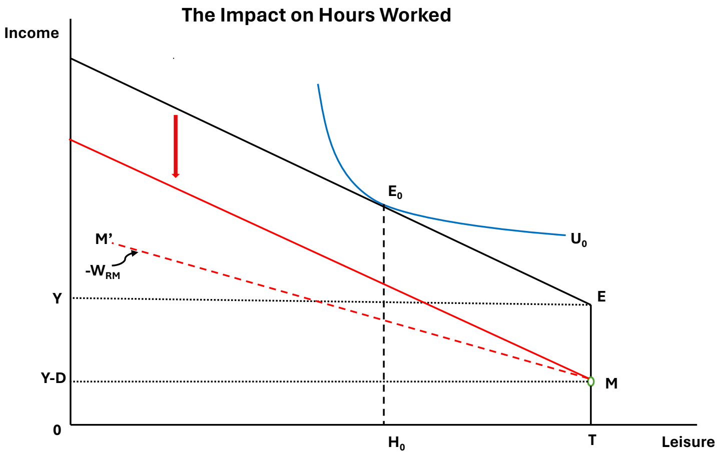

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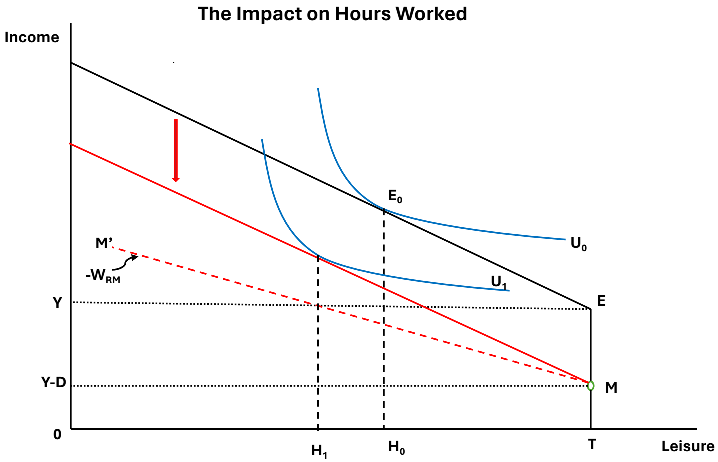

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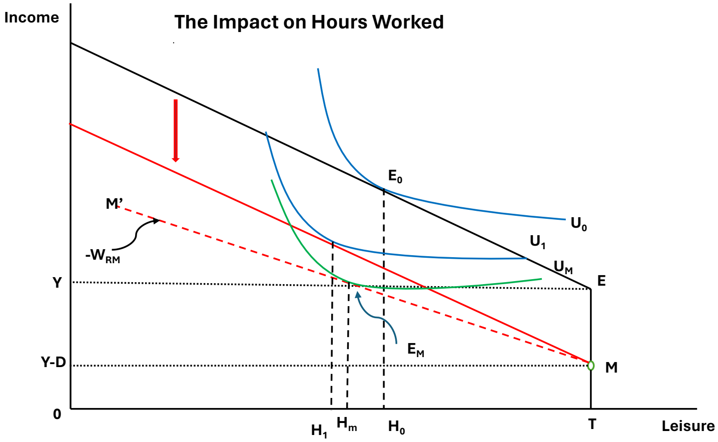

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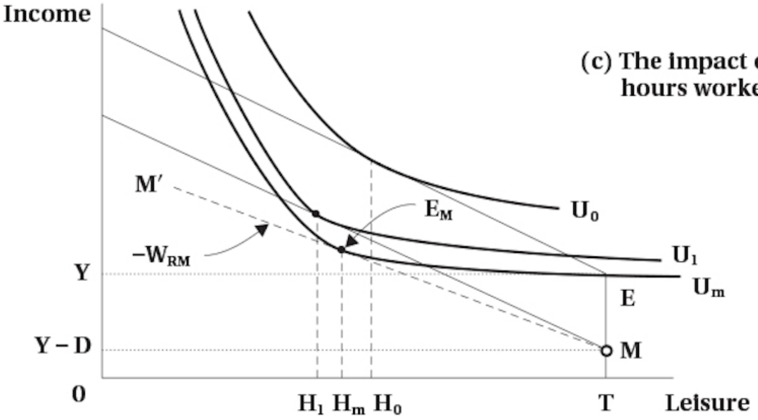

The effect of Child Care Costs on Labour Supply

The impact on hours worked:

- initial equilibrium: at E_0 with hours of work H_0 and no child care costs
- the existence of fixed child care costs shifts the budget constraint down in a parallel fashion because market wages have not changed
- if leisure is a normal good - this pure income effect from the reduced income increases hours of work from H_0 to H_1
- when there are fixed costs, one has to work more hours to make up for the income loss associated with the costs of child care.

The effect of Child Care Costs on Labour Supply

Fixed child care costs also create a discontinuity or gap in the implied labour supply schedule

- the distance TH_m indicates the number of hours below which it would not be worthwhile to enter the labour market - given the reservation wage W_{RM} created by the child care costs EM
- it is not possible to amortize the fixed costs of entering the labour force over so few hours $\rightarrow$ there is a discrete jump in the labour supply from zero to TH_m
- geometrically, it is not possible for there to be a tangency between the budget constraint MM' and the indifference curve U_M in the region $E_M E$
  - the slope of MM' is always greater than the slope of the indifference curve to the right E_M
  - in this way, the child care costs discourage part-time work

Child Care Benefit

- Effects of child care costs
  - Fixed child care costs increase reservation wage - decrease labour force participation
  - Fixed child care costs also affect hours-of-work
    - Increased hours for those who participate
    - Create a discontinuity in labour supply - making part-time work less likely

Therefore, child care benefit:

- encourages labour force participation and part-time work

- reduces the hours of work for those already participating

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- Income maintenance schemes top up low incomes from varied causes; some are better for transitory shocks, others for persistent low wages.
- The labour supply framework compares work incentives across designs via their budget constraints.
- **Demogrants:** Pure income effect; reduce work incentives at the margin.
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